

A

B

C

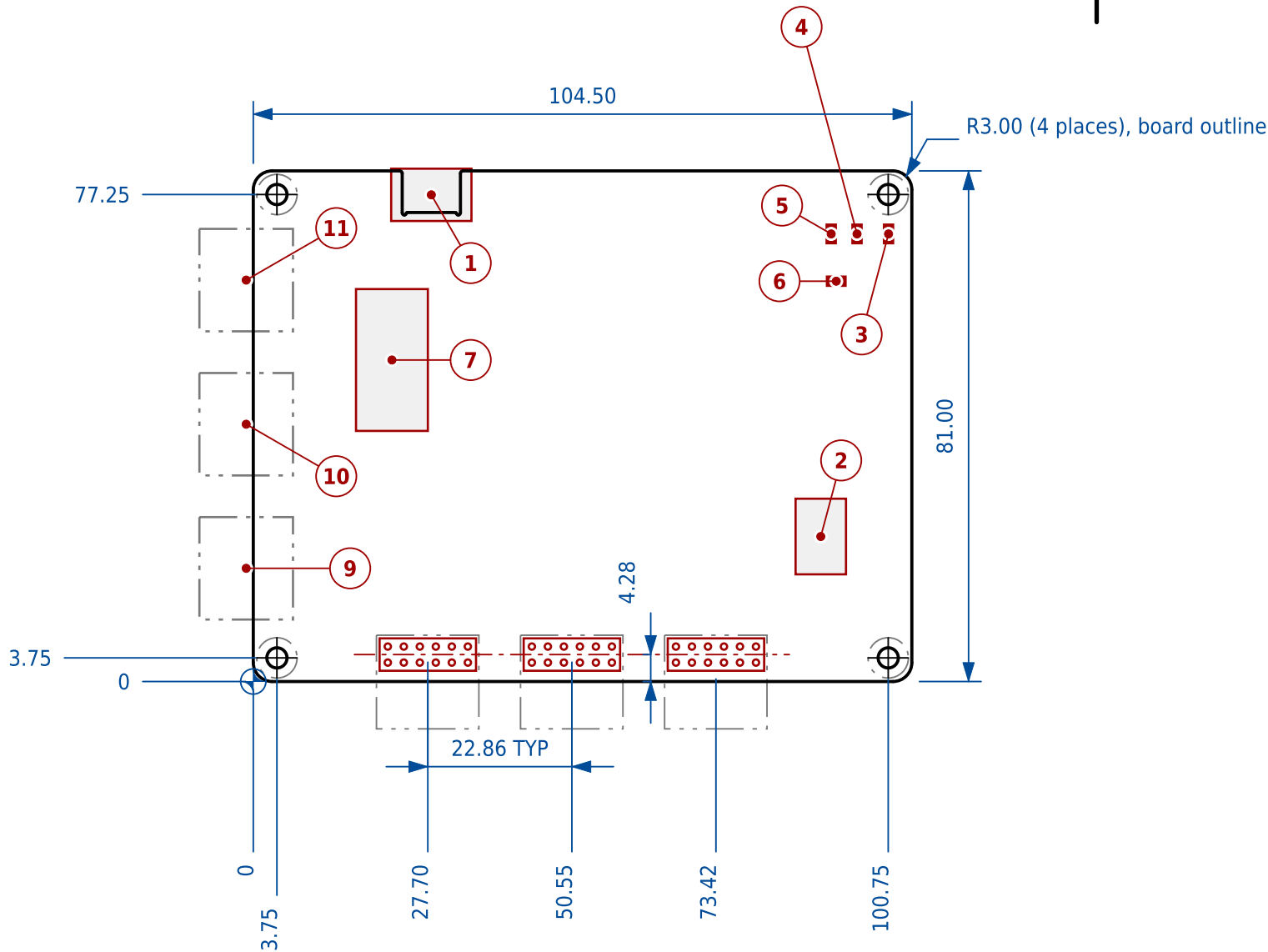
D

A

B

C

D



NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Ships in TT03p5 ASIC IC kit, TT04 Dev Kit and TT05 Dev Kit. A kit pairs this board with a breakout carrying the chip; the board itself is the same in each.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- The spreadsheet records 1.2.1 for TT03p5 and flags it unconfirmed: the TT03p5 render is labelled v1.1.2.
- v1.2.2c is the spreadsheet's name for 1.2.3: an assembly variant, no X1 and a 0R at R51.
- Covers revisions 1.2.1, 1.2.2 and 1.2.3, the same board mechanically, compared feature by feature before merging; only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.

NOTES (continued)

- Assembled envelope, connector overhang included, is 104.50 x 81.35 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- Upper edge carries a shallow recess for the USB-C shell, contributed by the connector footprint's own edge cuts.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ 8aad3f8 - title block: TinyTapeout 4+ Demoboard rev 1.2.1, dated 2024-04-12
- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ cfdd80d7b - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26
- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad_pcb @ a88cbc08b - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.75	3.75	3.20	6.40
MT2	100.75	3.75	3.20	6.40
MT3	3.75	77.25	3.20	6.40
MT4	100.75	77.25	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	27.695	4.275	34.045	5.545
BIDIR	J5	bottom	50.555	4.275	56.905	5.545
OUTPUT	J6	bottom	73.415	4.275	79.765	5.545

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	21.88 to 34.62	73.05 to 81.35
2	7-segment display	86.05 to 94.05	17.00 to 29.00
3	LED D1	100.07 to 101.53	69.52 to 72.48
4	LED D2	95.07 to 96.53	69.52 to 72.48
5	LED D3	90.97 to 92.43	69.52 to 72.48
6	LED D4	91.02 to 93.98	62.77 to 64.23
7	8-way input DIP switch	16.30 to 27.70	39.75 to 62.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-8.55 to 6.30	9.83 to 26.08
10	Side Pmod J13 (not fitted)	-8.55 to 6.30	32.69 to 48.94
11	Side Pmod J14 (not fitted)	-8.55 to 6.30	55.55 to 71.80

LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

SOURCES (continued)

- Shuttle mapping:
https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrfFzF0urjLVWMdqAnrOJLetM/edit?gid=1250963370 - used by: TT03p5, TT04, TT05

DB 4+ V1.2.1 / DB 4+ V1.2.2 / DB 4+ V1.2.2C

tinytapeout-demo rev 1.2.1, 1.2.2 and 1.2.3

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-103-g47e2c44	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-02	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	